RUDRA CHAUHAN

Computer Engineering Student
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PROFESSIONAL SUMMARY

Final-year Computer Engineering student building production-shaped full-stack, AI, and security systems — from Isolation Forest anomaly detection and LLM threat intelligence to OWASP-aligned prompt-injection defenses. Ships across React/TypeScript, FastAPI/Flask, and Dockerized cloud deployments, translating architecture decisions into working, deployed software.

CORE COMPETENCIES

Full-Stack Application Development · AI Application Engineering · Computer Vision · Secure Software Design · Cloud Deployment · REST API Development · Product Thinking

EDUCATION

B.E. Computer Engineering, Minor in Data Science — C.K. Pithawala College of Engineering & Technology (GTU), Surat Jul 2023 – 2027 (Expected)
CGPA: 8.22/10 | Currently in 7th Semester

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL
Frameworks: React, FastAPI, Flask, Tailwind CSS, Vite, Firebase
AI/ML: Computer Vision, LLMs (OpenAI/Ollama), Prompt Engineering, scikit-learn (Isolation Forest)
Cloud & DevOps: Docker, Docker Compose, Nginx, Vercel, Render
Security: OWASP LLM Top 10, MITRE ATT&CK, JWT/bcrypt, RBAC
Databases & Tools: PostgreSQL, Firebase (Auth/Firestore/Storage), Git, GitHub, Swagger, Postman

PROJECTS

CVision — Intelligent Resume Screening Platform — *React · TypeScript · Firebase · Vite* | GitHub | Demo

- Designed a recruiter-facing resume screening platform on a serverless Firebase backend (Auth, Firestore, Storage), isolating all Firebase logic behind a dedicated services layer and enforcing account-scoped data isolation via granular Firestore rules and route guards at both client and database layers.
- Built a deterministic, rule-based resume-parsing pipeline for skills and experience extraction, ensuring every parsed field traces to a visible rule for fully explainable candidate summaries.

CortexGuard — AI Security Operations Center — *FastAPI · React · PostgreSQL · Docker* | GitHub

- Architected a containerized React + FastAPI + PostgreSQL Security Operations Center with Docker Compose and Nginx, unifying 4 independent log sources (SSH, Apache, Nginx, firewall) into one real-time pipeline.
- Engineered a dual-layer detection engine — an 8-rule sliding-window system plus an Isolation Forest anomaly model — to catch both known and previously unseen attack patterns.
- Designed a 3-layer Prompt Firewall (16 regex interception rules + OWASP-aligned output-schema validation) that maps every alert to 1 of 18 MITRE ATT&CK techniques, secured behind JWT/bcrypt auth, RBAC, and per-IP rate limiting.

FrameGuard — Deepfake & AI-Generated Media Detection — *Flask · Python · Multimodal ML* | GitHub

- Built a multimodal computer vision system processing image, video, and audio through one forensic pipeline to detect deepfake and AI-generated media.
- Balanced cost and performance by separating an always-on lightweight cloud validation service from a compute-heavy local extraction pipeline, architected around free-tier cloud constraints.

PROFESSIONAL EXPERIENCE

Remarkskill Education — *Ethical Hacking Intern*Jun 2025 – Jul 2025

- Conducted hands-on vulnerability assessments under project supervision, applying structured methodology to identify and document security weaknesses.
- Authored a capstone vulnerability assessment report translating findings into prioritized, actionable remediation guidance.

YBI Foundation (Virtual) — *Python Programming for AI & Data Science Intern*2 Weeks, Jul 2026

- Completed hands-on training in Python programming, data handling, and analytics workflows for AI and data science applications.

HACKATHONS & ACHIEVEMENTS

Techmanthan 2026 (Winner) · Climate Data Hackathon 2026 (Finalist) · Hack & Forge 2026 (Participant)

CERTIFICATIONS

IBM Generative AI for Data Scientists (Coursera, 2026) | Verify Credential · **NPTEL Data Mining** (IIT Kharagpur, 2026)